Synthesis of Butyl Pelargonate via Fischer Esterification

Aidan Leonard, Dr. Kawagoe, Dr. MacPherson

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1 Introduction

The objective of this experiment is to synthesize butyl pelargonate via Fischer Esterification by reacting 1-butanol with pelargonic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

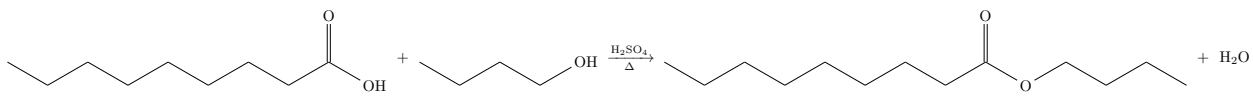


Figure 1: Reaction of Pelargonic Acid and 1-Butanol

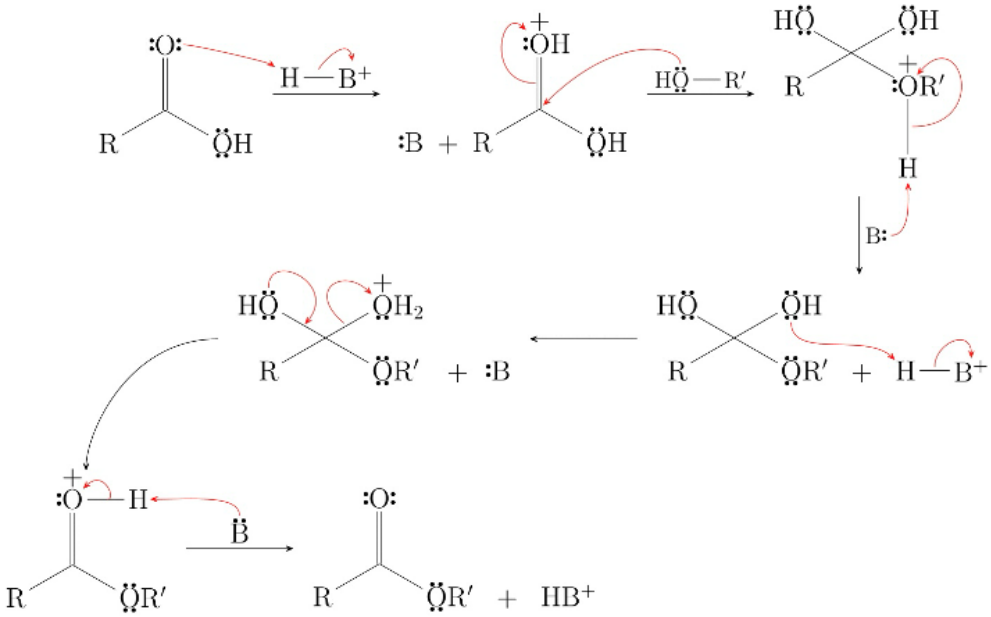


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Pelargonic Acid	158.24	0.903	254 - 256	1.0	4.784 g / 5.30 mL	30.2
1-Butanol	74.123	0.81	117.7	2.0	4.482 g / 5.53 mL	60.5
Sulfuric Acid	98.08	1.84	–	cat.	3 drops	–
Diethyl Ether	74.12	0.710	34.6	solv.	–	–
Butyl Pelargonate	214.34	0.864	258 - 260	–	6.480 g / 7.50 mL	30.2

4 Experimental Procedure

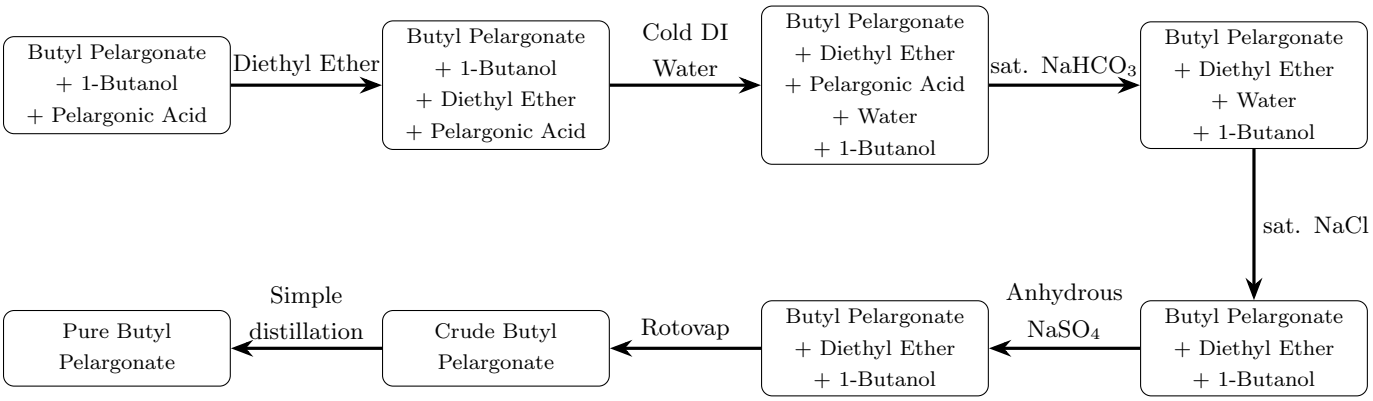
4.1 Apparatus

- Reflux
- Simple Distillation
- Rotovap

4.2 Procedure

- In a 25 mL round-bottom flask, add 4.784 g (5.30 mL) of pelargonic acid followed by 6.723 g (8.30 mL) of 1-butanol and a boiling bead.
- While stirring, slowly add 3 drops of concentrated sulfuric acid.
- Gently reflux for 60 minutes, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 10 mL total of diethyl ether to aid in the process.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this twice.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude butyl pelargonate.
- Purify the product with simple distillation.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Pelargonic acid	g of 1-Butanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product Observations	Purified Product Observations